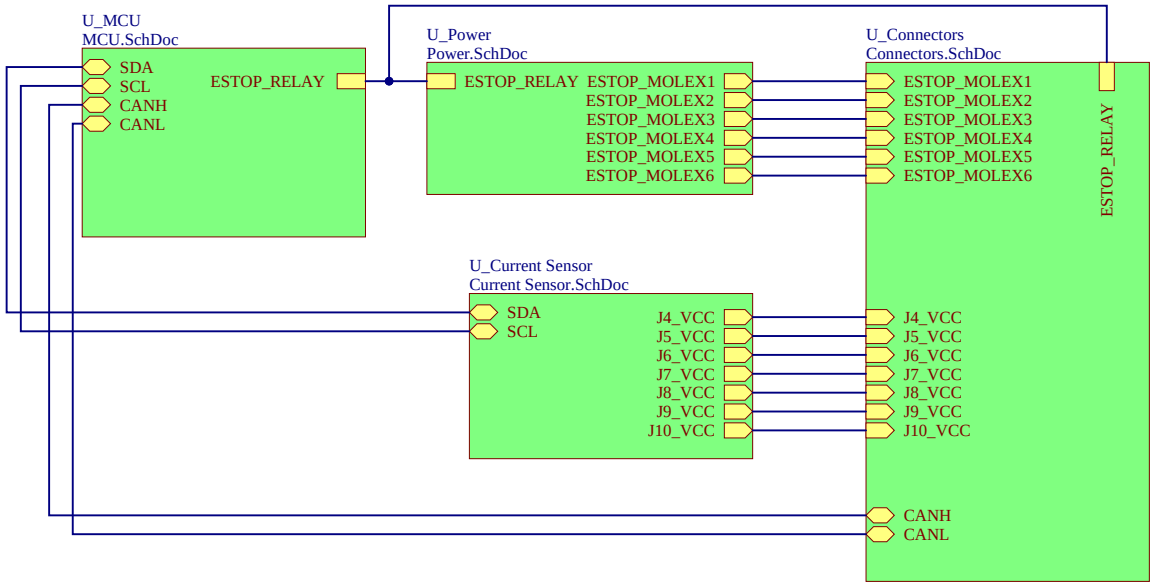
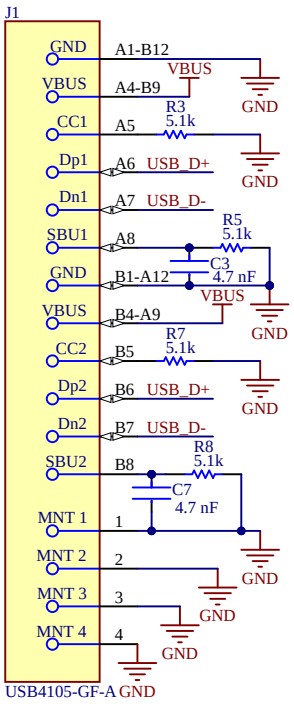
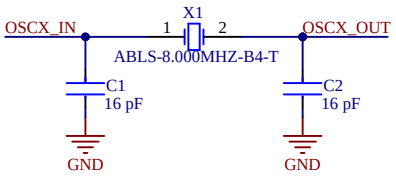


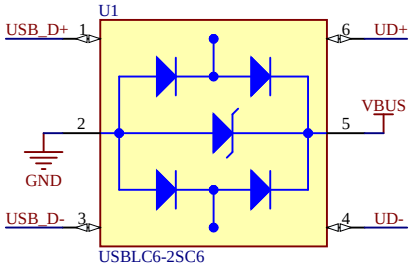
Title: **The Hub**			
Size: Letter	Drawn By: Sao Aphisith Sithisack		
Date: 11/10/2026	Revision: v1	Sheet 1 of 5	Team: IGVC

USB-C Connector

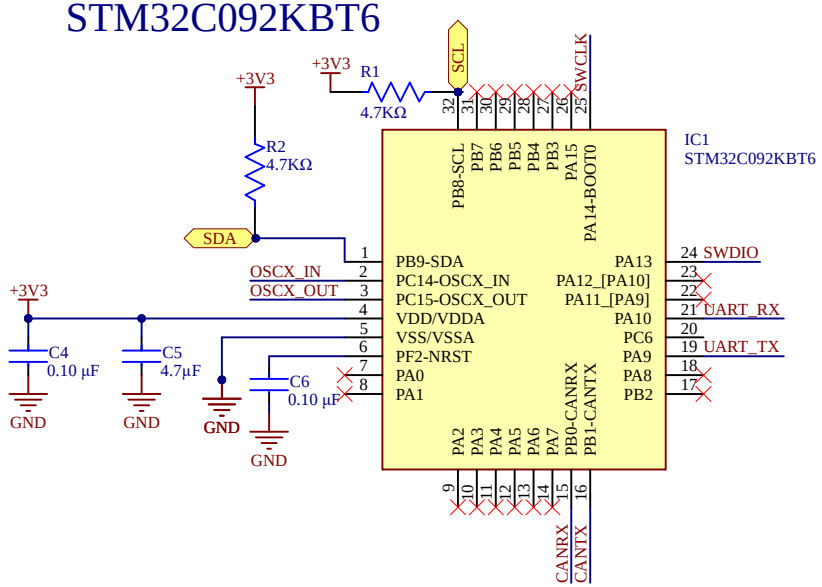


Crystal Oscillator

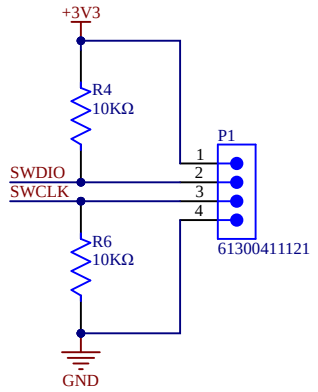




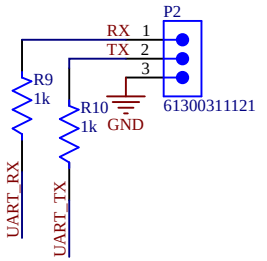
STM32C092KBT6



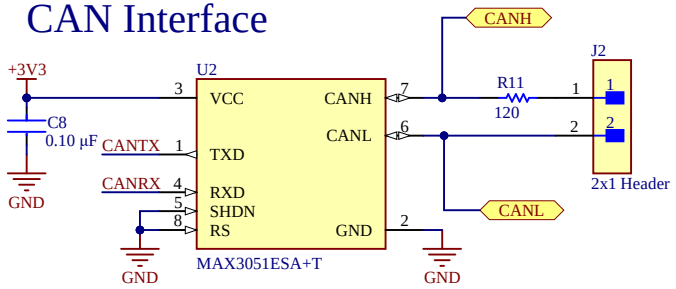
SWD Programming



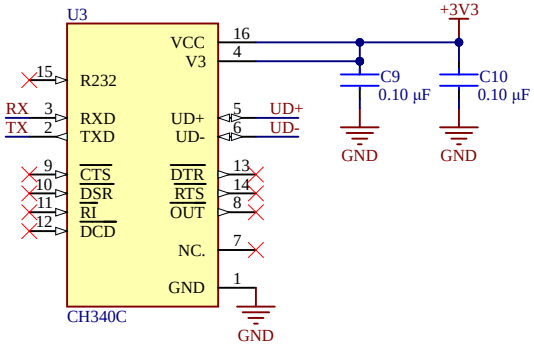
UART Programming



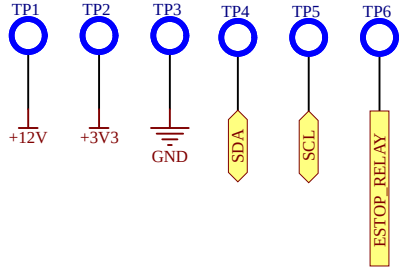
CAN Interface



USB-C to UART Bridge

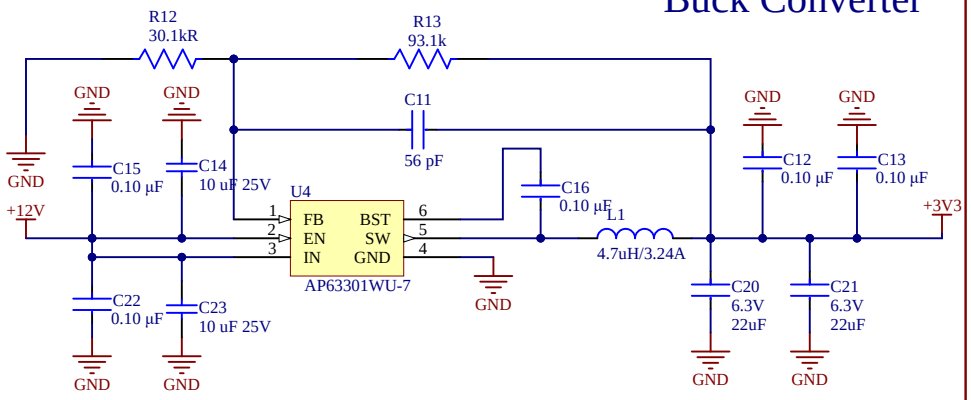


Test Points

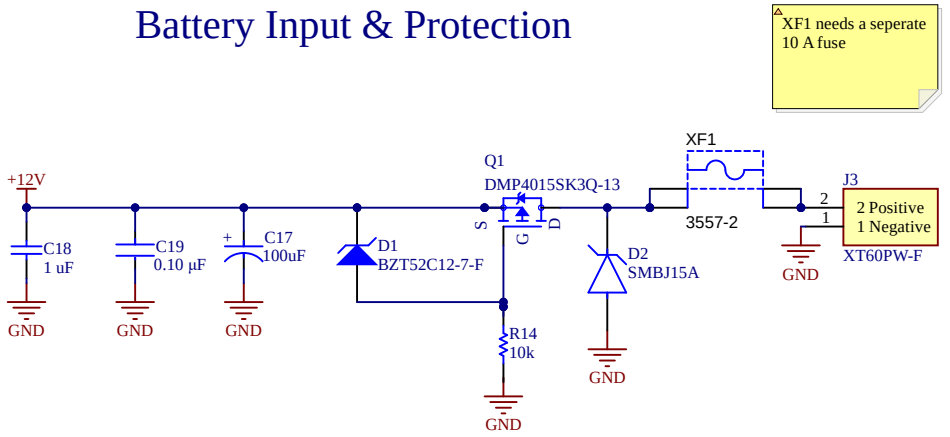


Title: **Hub MCU**			
Size: Letter	Drawn By: Sao Aphisith Sithisack		
Date: 11/10/2026	Revision: 1	Sheet 2 of 5	Team: IGVC

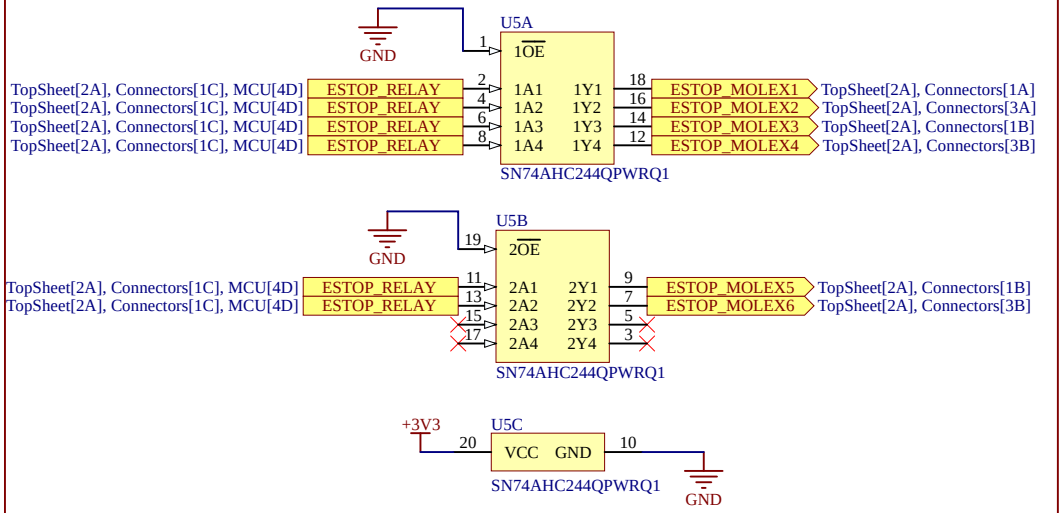
Buck Converter



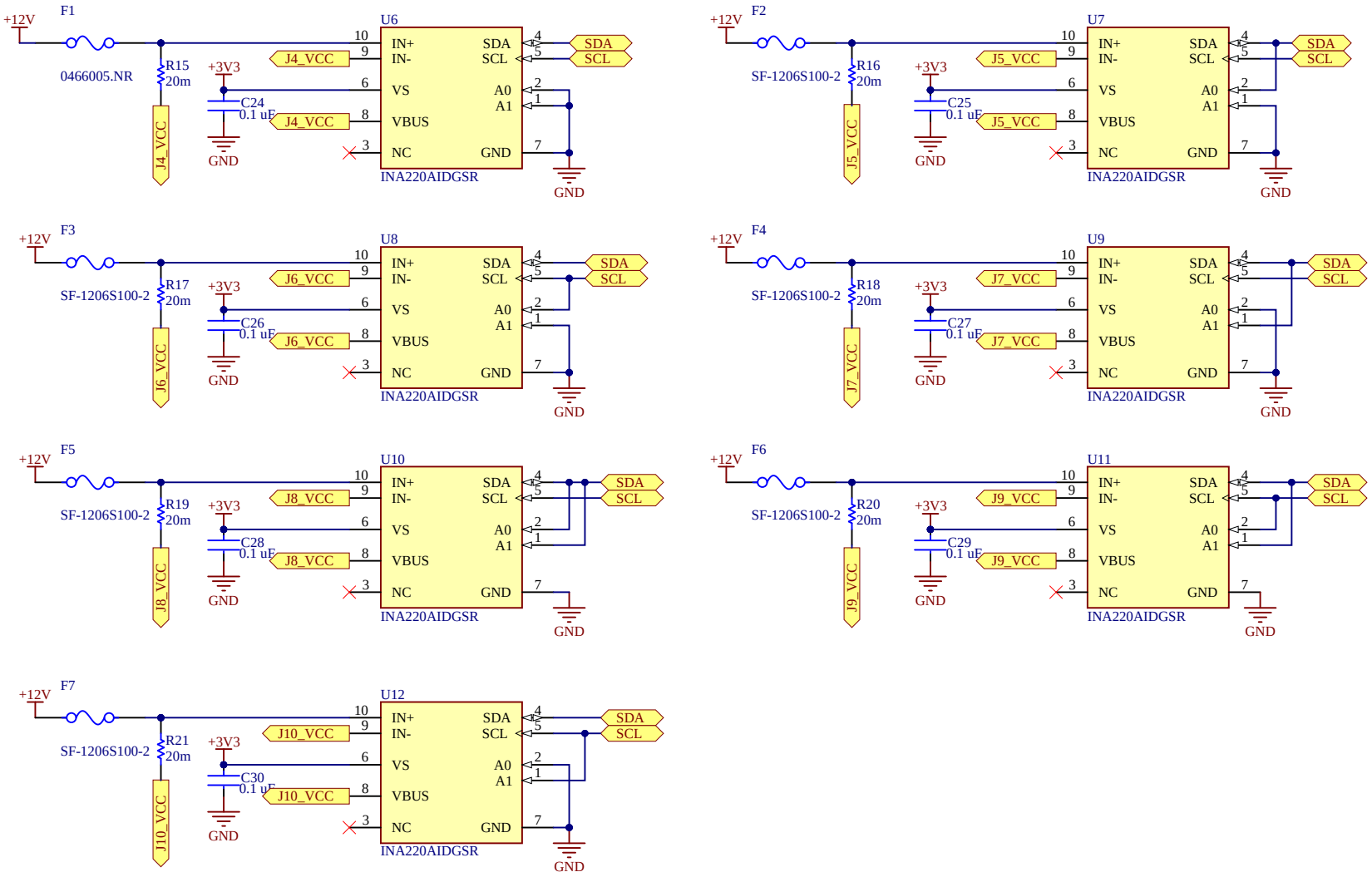
Battery Input & Protection



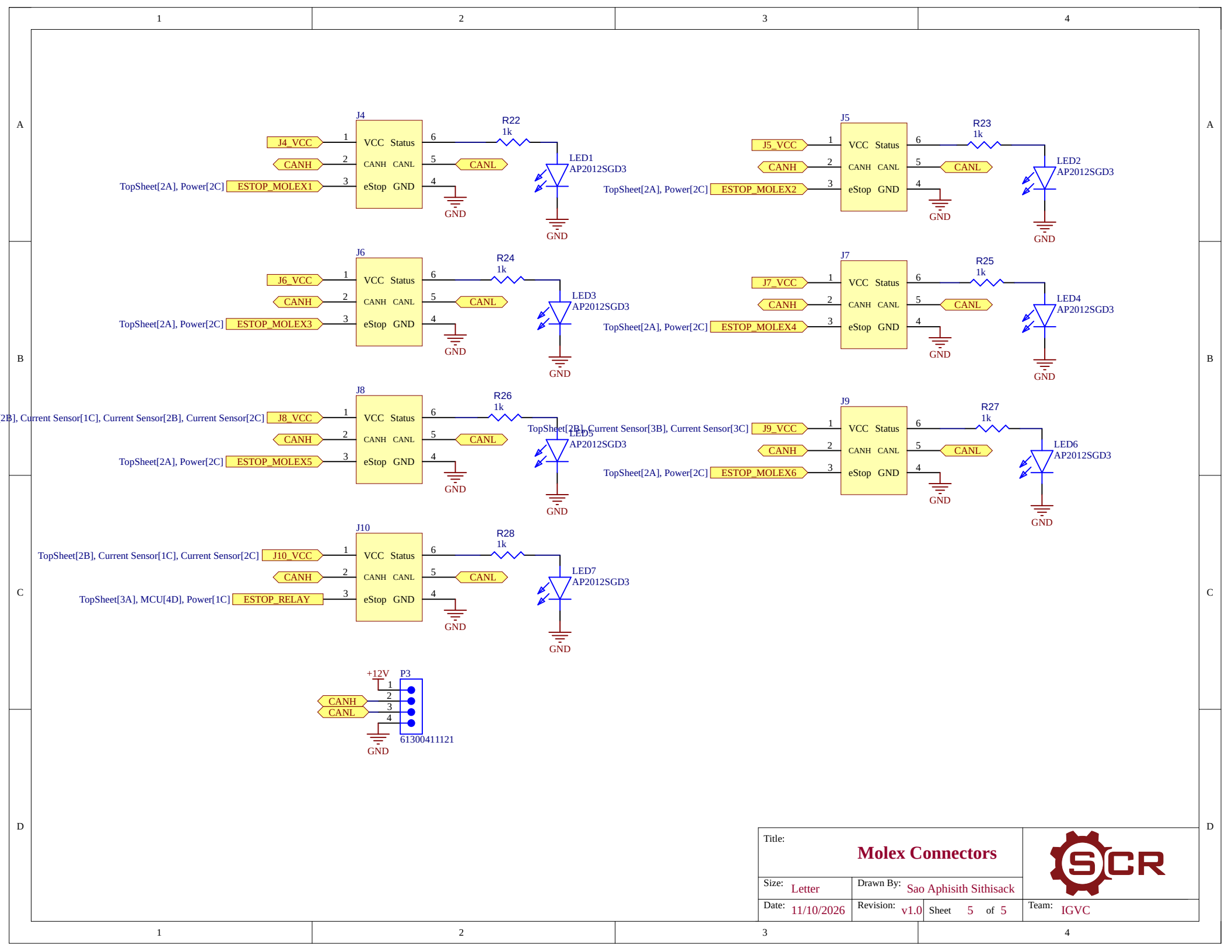
ESTOP Relay Buffers

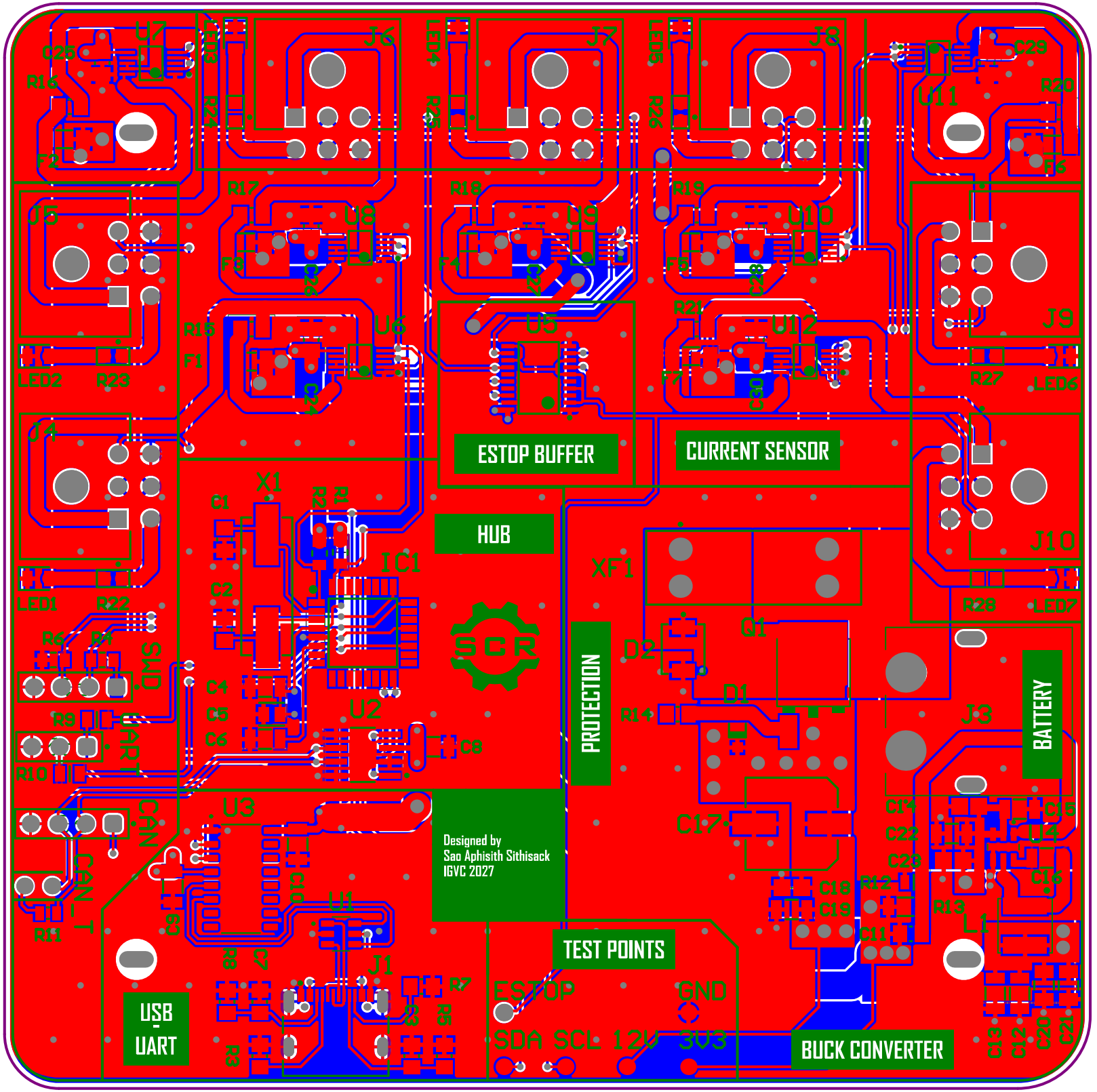


Title: **Hub Power**			
Size: Letter	Drawn By: Sao Aphisith Sithisack		
Date: 11/10/2026	Revision: 2.0	Sheet 3 of 5	Team: IGVC



Title: **Current Sensor**			
Size: Letter	Drawn By: Sao Aphisith Sithisack		
Date: 11/10/2026	Revision: v1.0	Sheet 4 of 5	Team: IGVC


J4
J4_VCC
CANH
TopSheet[2A], Power[2C]
ESTOP_MOLEX1
VCC Status
CANH CANL
eStop GND
CANL
GND
R22
1k
LED1
AP2012SGD3
J5
J5_VCC
CANH
TopSheet[2A], Power[2C]
ESTOP_MOLEX2
R23
1k
LED2
AP2012SGD3
J6
J6_VCC
CANH
TopSheet[2A], Power[2C]
ESTOP_MOLEX3
R24
1k
LED3
AP2012SGD3
J7
J7_VCC
CANH
TopSheet[2A], Power[2C]
ESTOP_MOLEX4
R25
1k
LED4
AP2012SGD3
J8
J8_VCC
CANH
TopSheet[2A], Power[2C]
ESTOP_MOLEX5
TopSheet[2B], Current Sensor[1C], Current Sensor[2B], Current Sensor[2C]
R26
1k
LED5
AP2012SGD3
J9
J9_VCC
CANH
TopSheet[2A], Power[2C]
ESTOP_MOLEX6
TopSheet[2B], Current Sensor[3B], Current Sensor[3C]
R27
1k
LED6
AP2012SGD3
J10
J10_VCC
CANH
TopSheet[3A], MCU[4D], Power[1C]
ESTOP_RELAY
TopSheet[2B], Current Sensor[1C], Current Sensor[2C]
R28
1k
LED7
AP2012SGD3
+12V
P3
CANH
CANL
61300411121
Title: Molex Connectors
Size: Letter
Drawn By: Sao Aphisith Sithisack
Date: 11/10/2026
Revision: v1.0
Sheet 5 of 5
Team: IGVC


ESTOP BUFFER
CURRENT SENSOR
HUB
PROTECTION
BATTERY
TEST POINTS
USB
UART
BUCK CONVERTER
Designed by
Sao Aphisith Sithisack
IGVC 2027
ESTOP
GND
SDA
SCL
12V
3V3